Xinyuan Song

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Ph.D. Student in Computer Science

Emory University

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2,507	24	51	September 2026
CITATIONS	H-INDEX	I10-INDEX	GOOGLE SCHOLAR

Education

- **Emory University** — 2024 – Present
  Ph.D. in Computer Science; focus on Machine Learning. Advisor: Liang Zhao.
- **University of Texas at Arlington** — 2023 – 2024
  Ph.D. studies in Computer Science. Advisor: Yingying Zhu.
- **National University of Singapore** — 2019 – 2023
  M.S. in Statistics. Advisor: Cheng Li.
- **Peking University** — 2014 – 2019
  B.S. in Physics; B.S. in Mathematics.

Industry Experience

- **Cielara AI** — Summer 2026
  Research on memory for language-model agents. MEMONDEMAND develops query-adaptive hierarchical memory for large enterprise collections; MegaMem provides source-resolved retrieval and attribution over ultra-large context windows.
- **Gradient Network** — Summer 2025
  Research on multi-agent systems, including GradientSys: A Multi-Agent LLM Scheduler with ReAct Orchestration.
- **Alibaba Group** — 2020 – 2021
  Full-time Machine Learning Engineer.

Selected Publications

1. CPAL · 2026 ORAL
   MASTERPIECE **Matrix Sensing with Kernel Optimal Loss: Robustness and Optimization Landscape**
   Xinyuan Song and Ziye Ma.
   PDF GITHUB BIBTEX
2. NeurIPS · 2025
   The Curse of Depth in Large Language Models
   Wenfang Sun*, **Xinyuan Song***, Pengxiang Li, Lu Yin, Yefeng Zheng, and Shiwei Liu.
   PDF GITHUB BIBTEX
3. NeurIPS · 2025
   TRiCo: Triadic Game-Theoretic Co-Training for Robust Semi-Supervised Learning
   Hongyang He*, **Xinyuan Song***, Yangfan He, Zeyu Zhang, Yanshu Li*, Haochen You, Lifan Sun, and Wenqiao Zhang.
   PDF BIBTEX
4. ICML · 2026
   Newton-Coupled Dual-Teacher Semi-Supervised Learning Framework
   Hongyang He*, **Xinyuan Song***, Yan Zhong, Daizong Liu, Xuanyu Liu, and Victor Sanchez.
   PDF BIBTEX

5. **ICML · 2026**
   When Does Sparsity Mitigate the Curse of Depth in LLMs
   Dilxat Muhtar*, **Xinyuan Song***, Sebastian Pokutta, Max Zimmer, Nico Pelleriti, Thomas Hofmann, and Shiwei Liu.
   PDF GITHUB BIBTEX
6. **ICASSP · 2026**
   Demystifying the Roles of LLM Layers in Retrieval, Knowledge, and Reasoning
   Xinyuan Song, Keyu Wang, Pengxiang Li, Lu Yin, and Shiwei Liu.
   PDF GITHUB BIBTEX
7. **CPAL · 2026**
   Efficient Temporal Consistency in Diffusion-Based Video Editing with Adaptor Modules: A Theoretical Framework
   Xinyuan Song, Yangfan He, Sida Li, Jianhui Wang, Hongyang He, Xinhang Yuan, Ruoyu Wang, Jiaqi Chen, Keqin Li, Kuan Lu, Menghao Huo, Binxu Li, and Pei Liu.
   PDF BIBTEX
8. **ACL Findings · 2026**
   P-QuASAR: A Unified Probabilistic Framework for Holistic Patent Quality Assessment and Refinement
   Xinyuan Song, Ziyi Ni, Fred Yang, Bo Zhang, Yiming Yan, Yijin Wang, and Jane Zhang.
   PDF BIBTEX
9. **CPAL · 2026**
   Token-Aware Representation Augmentation for Fine-Grained Semi-Supervised Learning
   Hongyang He*, Yan Zhong*, **Xinyuan Song***, Daizong Liu, and Victor Sanchez.
   PDF BIBTEX
10. **ECCV · 2026**
    Geometric Regularization for Long-Tailed Semi-Supervised Learning via Gaussian Feature Bridges
    Hongyang He*, **Xinyuan Song***, Yan Zhong, Daizong Liu, Yanbin Li, Yangfan He, and Wenqiao Zhang.
    PDF BIBTEX
11. **ACM Multimedia · 2025**
    Free-Mask: A Novel Paradigm of Integration Between the Segmentation Diffusion Model and Image Editing to Improve Segmentation Ability
    Bo Gao*, Jianhui Wang*, **Xinyuan Song***, Yangfan He, Fangxu Xing, and Tianyu Shi.
    PDF BIBTEX
12. **Heliyon · 2023**
    Predicting Stock Price of Construction Companies Using Weighted Ensemble Learning
    Xinyuan Song.
    PDF BIBTEX

Recent Study

1. **arXiv 2609.17419 · 2026**
   World Model Science: Self-Organized Criticality, Weak Chaos, and Metastable Belief Dynamics in Long-Horizon LLM Agents
   Xinyuan Song and Zekun Cai.
   PDF GITHUB BIBTEX
2. **arXiv 2608.22141 · 2026**
   MEMONDEMAND: A Memory Management System for Large-Scale Enterprise Data
   Xinyuan Song, Bowen Zhu, Hasibul Haque, and Liang Zhao.
   PDF GITHUB BIBTEX
3. **arXiv 2608.22137 · 2026**
   MegaMem: A Retrieval Solution for Ultra-Large Context Windows
   Xinyuan Song, Bowen Zhu, Hasibul Haque, and Liang Zhao.
   PDF GITHUB BIBTEX
4. **arXiv 2607.19331 · 2026**
   ISO: An RLVR-Native Optimization Stack
   Hanqing Zhu, Wenyan Cong, Zhizhou Sha, Sagnik Mukherjee, **Xinyuan Song**, David González-Martínez, Xiaoxia Wu, Yuandong Tian, Shiwei Liu, David Z. Pan, and Zhangyang “Atlas” Wang.
   PDF GITHUB BIBTEX
5. **arXiv · 2026**
   FedWorld: Scope-Aware Federation of Agent World Models
   Yuchao Hou, Guangyuan Dong, **Xinyuan Song**, Hangtao Zhang, Gaoyuan Du, Jiachen Luo, and Zhengshan Wang.
   PDF GITHUB BIBTEX
6. **arXiv · 2026**
   When Do Safe Agents Misgeneralize Across Surfaces? Invariant Safety State Abstraction
   Yuchao Hou, Guangyuan Dong, Hangtao Zhang, **Xinyuan Song**, Jiachen Luo, Gaoyuan Du, and Xianlong Wang.
   PDF GITHUB BIBTEX

7. arXiv 2607.05458 · 2026
   Learning to Control LLM Agent Harnesses with Offline Reinforcement Learning
   Haiwen Yi and **Xinyuan Song**.
   PDF GITHUB BIBTEX
8. arXiv 2607.04535 · 2026
   ManifoldFlow: SPD-Relaxed Stiefel Layers with Learnable Singular Spectrum
   Haiwen Yi and **Xinyuan Song**.
   PDF GITHUB BIBTEX
9. arXiv 2607.04528 · 2026
   Measuring Harness-Induced Belief Divergence in Multi-Step LLM Agents
   Haiwen Yi and **Xinyuan Song**.
   PDF GITHUB BIBTEX
10. arXiv 2607.01767 · 2026
    Repair the Amplifier, Not the Symptom: Stable World-Model Correction for Agent Rollouts
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
11. arXiv 2607.00062 · 2026
    AlgoBench: Benchmarking Algorithmic Adaptation in Code Generation
    Xinyuan Song, Zekun Cai, and Liang Zhao.
    PDF GITHUB BIBTEX
12. arXiv 2606.31422 · 2026
    Ask the World Before Acting: Environment Probing for Calibrated Agent World Models
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
13. arXiv 2606.31399 · 2026
    World-Model Collapse as a Phase Transition
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
14. arXiv 2606.29999 · 2026
    AlgoSkill: Learning to Design Algorithms by Scheduling Human-Like Skills
    Xinyuan Song, Zekun Cai, and Liang Zhao.
    PDF GITHUB BIBTEX
15. arXiv 2606.28438 · 2026
    When AI Reviews Its Own Code: Recursive Self-Training Collapse in Code LLMs
    Xinyuan Song, Zekun Cai, and Liang Zhao.
    PDF GITHUB BIBTEX
16. arXiv 2606.27806 · 2026
    Agent vs. Parametric World Models: Hybrid Planning for Reliable Language Agents
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
17. arXiv 2606.27780 · 2026
    Understanding Rollout Error in Graph World Models
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
18. arXiv 2605.13716 · 2026
    SkillOps: Managing LLM Agent Skill Libraries as Self-Maintaining Software Ecosystems
    Xinyuan Song, Hongji Pu, and Liang Zhao.
    PDF GITHUB BIBTEX

Collaborative Publications

1. ACL Findings · 2026
   EffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling
   Jiacheng Shi, Hongfei Du, **Xinyuan Song**, Y. Alicia Hong, Yanfu Zhang, and Ashley Gao.
   PDF BIBTEX
2. CVPR · 2026
   DICArt: Advancing Category-level Articulated Object Pose Estimation in Discrete State-Spaces
   Li Zhang, Mingyu Mei, Ailing Wang, Xianhui Meng, Yan Zhong, **Xinyuan Song**, Liu Liu, Rujing Wang, Zaixing He, and Cewu Lu.
   PDF BIBTEX
3. ACM Multimedia · 2026
   Generalizable Blind Image Quality Assessment with Retrieval Augmentation and Contrastive Vision-Text Local Meta-Training
   Yan Zhong, Zhiyuan You, Xinping Zhao, Li Zhang, **Xinyuan Song**, Ruoyu Zhao, Lei Shi, and Tingting Jiang.

4. **CPAL · 2026**
   ShapLoRA: Allocation of Low-rank Adaptation on Large Language Models via Shapley Value Inspired Importance Estimation
   Colin Zhao, Qinghua Yao, **Xinyuan Song**, and Wei Zhu.
   PDF **BIBTEX**
5. **CPAL · 2026**
   Enhancing Low-Cost Video Editing with Lightweight Adaptors and Temporal-Aware Inversion
   Yangfan He, Sida Li, Jianhui Wang, Kun Li, **Xinyuan Song**, Xinhang Yuan, Keqin Li, Kuan Lu, Menghao Huo, Jingqun Tang, Yi Xin, Jiaqi Chen, Miao Zhang, and Xueqian Wang.
   PDF **BIBTEX**
6. **CPAL · 2026**
   Improving Medical Visual Reinforcement Fine-Tuning via Perception and Reasoning Augmentation
   Guangjing Yang, ZhangYuan Yu, Ziyuan Qin, **Xinyuan Song**, Huahui Yi, Qingbo Kang, Jun Gao, Yiyue Li, Chenlin Du, and Qicheng Lao.
   PDF **BIBTEX**
7. **ICASSP · 2026**
   Optimizing Multi-Round Enhanced Training in Diffusion Models for Improved Preference Understanding
   Kun Li, Jianhui Wang, Yangfan He, **Xinyuan Song**, Ruoyu Wang, Hongyang He, Wenxin Zhang, Jiaqi Chen, Keqin Li, Sida Li, Miao Zhang, Tianyu Shi, and Xueqian Wang.
   PDF **BIBTEX**
8. **ICASSP · 2026**
   Self-evolving Agents with Reflective and Memory-Augmented Abilities
   Xuechen Liang, Yangfan He, Yinghui Xia, **Xinyuan Song**, Jianhui Wang, Meiling Tao, Li Sun, Xinhang Yuan, Jiayi Su, Keqin Li, Jiaqi Chen, Jinsong Yang, Siyuan Chen, and Tianyu Shi.
   PDF **GITHUB** **BIBTEX**
9. **ICLR CAO Workshop · 2026**
   Manifold-Aware Temporal Domain Generalization for Large Language Models
   Yiheng Yao, Zekun Cai, **Xinyuan Song**, Hiroki Hill Kobayashi, Xuan Song, Ryosuke Shibasaki, and Liang Zhao.
   PDF **BIBTEX**
10. **ICLR LLM Reasoning Workshop · 2026**
    Enhancing Code LLMs with Reinforcement Learning in Code Generation: A Survey
    Junqiao Wang, Zeng Zhang, Yangfan He, Zihao Zhang, **Xinyuan Song**, Yuyang Song, Tianyu Shi, Yuchen Li, Hengyuan Xu, Kunyu Wu, Xin Yi, Zhongwei Wan, Xinhang Yuan, Zijun Wang, Kuan Lu, Menghao Huo, Tang Jingqun, Guangwu Qian, Keqin Li, Qiuwu Chen, and Lewei He.
    PDF **BIBTEX**
11. **WWW LARS Workshop · 2026**
    SCORE: Story Coherence and Retrieval Enhancement for AI Narratives
    Qiang Yi, Yangfan He, Jianhui Wang, **Xinyuan Song**, ShiYao Qian, Xinhang Yuan, Yi Xin, Yijin Wang, Jingqun Tang, Yuchen Li, Junjiang Lin, Hongyang He, Zhen Tian, Tianxiang Xu, Keqin Li, Kuan Lu, Menghao Huo, Jiaqi Chen, Miao Zhang, Tianyu Shi, and Jianyuan Ni.
    PDF **BIBTEX**
12. **CVPR DataCV Workshop · 2026**
    Beyond Data Augmentation: Energy-Based Kuramoto Neurons for Semi-Supervised Learning
    Yundi Hong, Ao Li, **Xinyuan Song**, Yan Zhong, Hongyang He, and Victor Sanchez.
    PDF **BIBTEX**
13. **BIO Integration · 2026**
    Large Language Models and Cognitive Science: A Comprehensive Review of Similarities, Differences, and Challenges
    Qian Niu, Junyu Liu, Ziqian Bi, Pohsun Feng, Benji Peng, Keyu Chen, Ming Li, Lawrence KQ Yan, Yichao Zhang, Caitlyn Heqi Yin, Cheng Fei, Tianyang Wang, Yunze Wang, Silin Chen, Ming Liu, Ziyuan Qin, Riyang Bao, **Xinyuan Song**, and Zekun Jiang.
    PDF **BIBTEX**
14. **NeurIPS · 2025**
    GPAS: Accelerating Convergence of LLM Pretraining via Gradient-Preserving Activation Scaling
    Tianhao Chen, Xin Xu, Zijing Liu, Pengxiang Li, **Xinyuan Song**, Ajay Kumar Jaiswal, Fan Zhang, Jishan Hu, Yang Wang, Hao Chen, Shizhe Diao, Shiwei Liu, Yu Li, Lu Yin, and Can Yang.
    PDF **GITHUB** **BIBTEX**
15. **TKDD · 2025**
    Bridging NIP and MLM: A Unified Meta-Learning Framework for Personalized Sequential Recommendation
    Youhua Li, Ersheng Ni, Yihao Liu, Sibo Xu, Tianyi Xu, Mingxuan Wu, Junchen Fu, Yucheng Zhang, Yuanqi He, **Xinyuan Song**, and Yongxin Ni.
    PDF **BIBTEX**
16. **EMNLP · 2025**
    SOLAR: Serendipity Optimized Language Model Aligned for Recommendation
    Zichen Yuan, Lifan Sun, Yucen Zhuang, Yue Wang, **Xinyuan Song**, Tianqi Xu, Siyuan Li, Junchen Fu, Youhua Li, Sirui Hong, Jiaqi Chen, Joemon M. Jose, and Yongxin Ni.
    PDF **GITHUB** **BIBTEX**

17. **CIKM · 2025**
AMBER: Adaptive Meta Balanced Paradigm for Heterogeneous Graph-Based Knowledge Tracing
Lifan Sun, Zichen Yuan, Ersheng Ni, Weihua Cheng, **Xinyuan Song**, Linkun Dai, Hongwei Jiang, Sibo Xu, Mengmeng Chen, Yucen Zhuang, Yongxin Ni, and Youhua Li.
PDF **BIBTEX**

18. **ACM Multimedia · 2025**
Adaptive Prompt Learning for Blind Image Quality Assessment with Multi-modal Mixed-datasets Training
Yan Zhong, Xinping Zhao, Li Zhang, **Xinyuan Song**, and Tingting Jiang.
PDF **BIBTEX**

19. **ICME · 2025**
ContrastiveGaussian: High-Fidelity 3D Generation with Contrastive Learning and Gaussian Splatting
Junbang Liu, Enpei Huang, Dongxing Mao, Hui Zhang, **Xinyuan Song**, and Yongxin Ni.
PDF **GITHUB** **BIBTEX**

20. **IEEE RA-L · 2025**
Towards the Integrated Category-level Articulation Perception and Manipulation Framework for Embodied Intelligence
Li Zhang, Wenbo Xu, Jiankun Wang, **Xinyuan Song**, Qi Wu, Rujing Wang, Xue Wang, and Liu Liu.

21. **Neurocomputing · 2025**
Enhancing Intent Understanding for Ambiguous Prompt: A Human-Machine Co-adaption Strategy
Yangfan He, Jianhui Wang, Yijin Wang, Yan Zhong, **Xinyuan Song**, Junjiang Lin, Xinhang Yuan, Jingqun Tang, Yi Xin, Hao Zhang, Yuchen Li, Zijian Zhang, Hongyang He, Tianxiang Xu, Miao Zhang, Kuan Lu, Menghao Huo, Keqin Li, Jiaqi Chen, Tianyu Shi, and Jianyuan Ni.
PDF **BIBTEX**

22. **Transportation Research Record · 2025**
Who Can You Trust on the Road? Adversarial Intentions in Cooperative MARL
Siyang Zhang, Yecheng Lyu, Luyang Gong, **Xinyuan Song**, and Tianyu Shi.
PDF **BIBTEX**

23. **BIBM · 2025**
Multi-Scale Volumetric Transformers with Adaptive Uncertainty Modeling for Robust Bacterial Flagellar Motor Localization in Cryo-Electron Tomography
Tianxiang Xu, Junqiao Wang, Yangfan He, **Xinyuan Song**, Peilai Yu, Xunfei Zhu, and Chang Liu.
PDF **BIBTEX**

24. **BIO Integration · 2025**
From Bench to Bedside: A Review of Clinical Trials in Drug Discovery and Development
Tianyang Wang, Ming Liu, Benji Peng, **Xinyuan Song**, Charles Zhang, Xintian Sun, Qian Niu, Junyu Liu, Silin Chen, Keyu Chen, Ming Li, Pohsun Feng, Ziqian Bi, Yunze Wang, Yichao Zhang, Cheng Fei, Lawrence KQ Yan, Ziyuan Qin, Riyang Bao, and Zekun Jiang.
PDF **BIBTEX**

25. **Neural Networks · 2024**
Semi-Supervised Multi-Label Feature Selection with Consistent Sparse Graph Learning
Yan Zhong, Xingyu Wu, Xinping Zhao, Li Zhang, **Xinyuan Song**, Lei Shi, and Bingbing Jiang.
PDF **BIBTEX**

* Equal contribution.

Honors and Awards

- Silver Medal, Chinese Physics Olympiad (national level)
- Silver Medal, Chinese Mathematics Olympiad (national level)